

OMAR MOHAMED ABDELSALAM

Digital IC Design & Verification | RTL • Verification • ASIC/FPGA Flow

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EDUCATION

Bachelor of Science in Electronics & Communication Engineering | *Helwan University, Cairo, Egypt* 2022 – 2027

TECHNICAL SKILLS

HDL / Languages: Verilog, System Verilog, C/C++, TCL

EDA Tools: Siemens Questa Sim, Questa Lint, Synopsys Design Compiler, Synopsys Spyglass (Lint/CDC/RDC), Xilinx Vivado

Domains: RTL Design, ASIC RTL-to-GDS Flow, Static Timing Analysis (STA), Clock Domain Crossing (CDC), Design-for-Test (DFT), Low-Power Design, Formal Verification, Gate-Level Simulation

Verification: System Verilog OOP, Assertions (SVA), Functional Coverage, Constrained-Random Verification

PROJECTS

UART (TX & RX) | *Verilog, Questa Sim, Spyglass*

- Designed and implemented a UART transmitter and receiver in Verilog supporting configurable parity (even/odd/disabled), busy-flag control, and serial frame generation per specification.
- Built a self-checking testbench validating frame transmission, parity generation, reset behavior, and busy logic at a 200 MHz simulation clock; ran RTL lint checks in SpyGlass to catch structural issues pre-simulation.
- Run Top module on Artix-7 FPGA.

Parameterized Synchronous FIFO | *Verilog, Spyglass, Questa Sim*

- Designed a width/depth-configurable FIFO using a circular-buffer architecture with independent read/write pointers and full/empty flag generation.
- Verified corner cases — write-to-full and read-from-empty — through targeted simulation and RTL linting in Spyglass.

SPI Slave with Single-Port RAM | *Verilog, FSM*

- Designed an FSM-based SPI slave interfaced to a single-port RAM.
- Compared Gray, one-hot, and sequential state encodings and selected the optimal scheme based on timing analysis results.

Spartan-6 DSP48A1 Slice | *Verilog, Xilinx Vivado, Questa Sim*

- Designed a Spartan-6 DSP48A1 slice: arithmetic Datapath, configurable registers, cascade paths, and carry logic.
- Implemented OPMODE-controlled functionality and validated correctness through simulation in Questa Sim.

INTERNSHIPS & COURSES

Digital IC Design Diploma | *Eng. Ali El-Temseh*

Jun. 2026 – Sep. 2026

- Trained across the full front-end-to-back-end ASIC flow: Verilog RTL design, self-checking test benches, Tcl scripting, STA, low-power design, CDC, Spyglass (Lint/CDC/RDC), Synopsys Design Compiler, DFT, formal verification, and gate-level simulation.
- Capstone project: RTL-to-GDS implementation of a configurable multi-clock, low-power digital system — designed and integrated UART TX/RX, an ALU, a register file, a FIFO, a clock divider, synchronizers, clock gating, and the top-level controller.

Digital Verification Using System Verilog & UVM | *Eng. Kareem Waseem*

Mar. 2026 – Apr. 2026

- Trained in System Verilog OOP, interfaces, assertions (SVA), functional coverage, and constrained-random verification, plus UVM methodology — sequences, TLM, phasing, and factory pattern.
- Applied simulation-based verification techniques, wrote verification plans, and ran code-coverage analysis; gained exposure to formal verification, FPGA prototyping, and emulation.

Digital Design Using Verilog & FPGA Flow | *Eng. Kareem Waseem*

Feb. 2026 – Mar. 2026

- Designed combinational and sequential RTL circuits, FSMs, and memory architecture in Verilog.
- Built self-checking testbenches with Tcl-automated simulation in Siemens Questa Sim; implemented and validated designs on a Xilinx Vivado / Basys 3 FPGA flow with STA and IP integration.

Sales & Business Development Intern | *Schneider Electric Egypt*

Aug. 2025 – Sep. 2025

- Gained exposure to EcoStruxure, energy management, and industrial automation solutions; supported sales channels on customer-facing engineering solutions.